

WASP-39b

R-1067

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_250403 · created 2026-07-30T12:59:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460768.942 +/- 0.015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.024
Transit depth (Rp/Rs)^2	2.56 +/- 0.75 [%]
Orbital Inclination (inc)	89.7 +/- 1.22
Airmass coefficient 1 (a1)	0.88 +/- 0.014
Airmass coefficient 2 (a2)	0.1 +/- 0.19
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#2 - [253, 380]
Optimal Aperture	3.36
Optimal Annulus	16.67
Transit Duration (day)	0.067 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.024 vs 0.1457 (deviation 0.6σ)
Duration fit vs published	0.067 d vs 0.12 d (deviation 3.53σ)
Mid-time O–C	10.7 min
Depth SNR	6.7
Residual scatter	1.64 %

Light curve

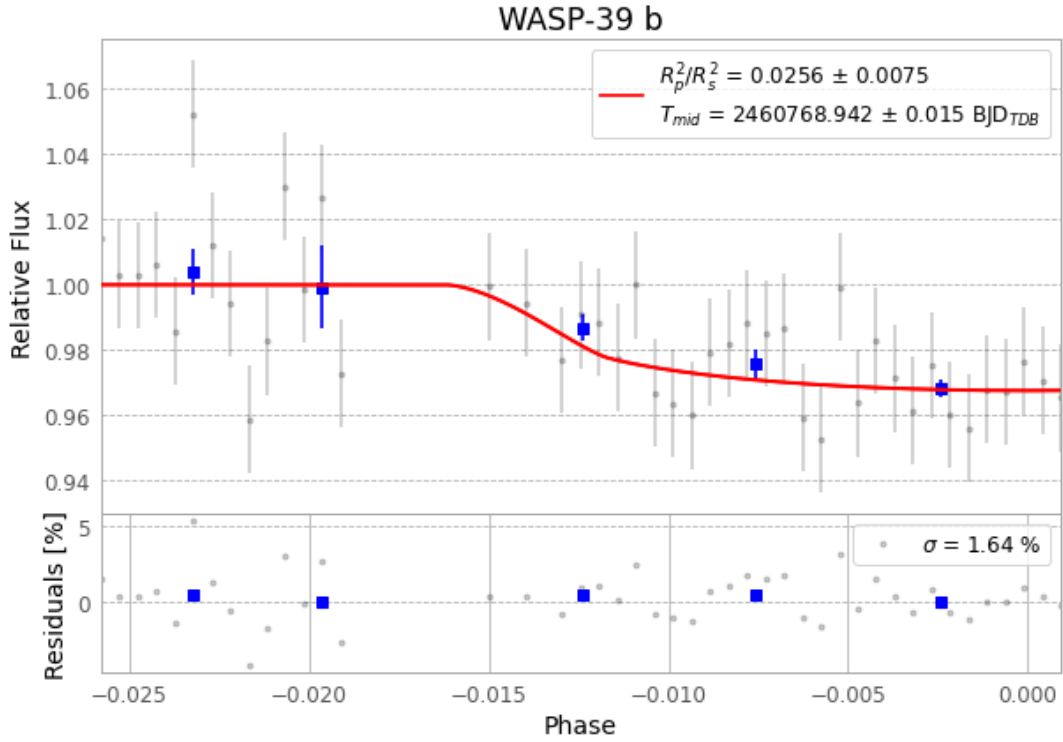


Figure 1 — FinalLightCurve_WASP-39 b_2025-04-03.png (SHA-256 c9fe5ae98447017...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [343, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 277bf7226298586f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

53 FITS frames (35 MB), WASP-39250403075716.FITS → WASP-39250403103315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-250403.log (4dc264ef85e3...), FinalLightCurve_WASP-39 b_2025-04-03.png (c9fe5ae9844...), FinalLightCurve_WASP-39 b_2025-04-03.csv (17ba8d6c67a3...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 12:59Z.